Forensic Analysis of I2P Communication Network in Android and macOS Environments

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***Abstract*— The Invisible Internet Project (I2P) has emerged as a crucial privacy-centric communication network, offering users a secure and anonymous online environment. As the utilization of I2P continues to expand, the demand for specialized forensic techniques tailored to this network becomes imperative. This paper comprehensively investigates I2P forensics within Android and macOS environments, addressing the unique challenges inherent in extracting and scrutinizing digital evidence from I2P applications. The research explores various artifacts and techniques to aid in reconstructing user activities within the I2P network, including analyzing storage files, configuration data, logs, and cache files. Additionally, the study investigates memory forensics as a complementary approach to I2P forensics. The outcomes of this research contribute to advancing forensic analysis techniques in privacy-centric networks, facilitating the efficient investigation of I2P-related crimes.**

Keywords*— *I2P forensics; privacy-centric networks; digital evidence; Android forensics; macOS forensics; memory forensics; dark web; digital forensics; eepsite, Anti-forensics.

I. Introduction

The pervasive integration of digital devices, including laptops, smartphones, tablets, and workstations, has fundamentally transformed numerous facets of contemporary existence. Based on Statista [1], as of the beginning of 2023, approximately 6.4 billion individuals worldwide, accounting for 64.4 percent of the global population, utilize the Internet, as shown in Fig. 1. These statistics underscore digital devices' integral role in modern lifestyles, with expectations of continued growth.

These devices operate on diverse operating systems, including Android, iOS, Bada, Windows OS, and Mac OS. Market shares reveal that Android leads with 34.78 percent, followed by Windows with 29.28 percent, iOS with 16.94 percent, OS X with 8.93 percent, and other systems. However, this proliferation of technologies also brings forth a simultaneous increase in cybercriminal activities, presenting considerable challenges for law enforcement agencies attempting to navigate the ever-evolving cyberspace. Individuals increasingly resort to anonymous communication methods to counteract censorship, surveillance, and other cyber-related concerns [2]. Some users strengthen their anonymity by combining anonymity applications with virtual private network (VPN) services, complicating investigations into such communications. Amid this landscape, the Invisible Internet Project (I2P) has emerged as a prominent anonymity network prioritizing privacy and security by design. I2P is embraced by lawful users and cyber criminals for various motives [3].

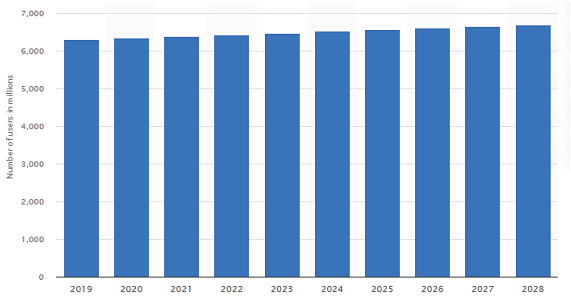

Fig. 1. Internet users' statistics [1]

It facilitates evading censorship, preserving privacy, exercising freedom of speech, participating in activism, accessing restricted content, and addressing security apprehensions [4], [28], [33].

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I2P's architecture is built upon layers of encryption, end-to-end data encryption, and routing data through a decentralized network of peers to access hidden services. However, investigating I2P communications poses complex challenges for law enforcement agencies. Prominent researchers have worked in this field [34] – [36].

Nonetheless, by methodically examining devices of interest, such as seized mobile phones or PCs, traces of illicit online activities can be unearthed, streamlining the investigation process [29], [37]. The motivation for this study stems from the escalating complexity of digital investigations catalyzed by advancements in operating systems and application development technologies. While existing research has delved into I2P forensics within the Windows operating system context, there is lack of exploration in Android and macOS. Remarkably, no research on the latest macOS and Android 11 concerning I2P communication has been conducted.

This paper presents a comprehensive study on I2P forensics in Android and macOS environments, addressing the challenges of extracting and analyzing digital evidence from I2P applications. The research simulates a dark web cybercrime scenario and scrutinize I2P communication artifacts on the macOS and Android operating systems to unveil suspects' online activities. Additionally, the research investigates techniques for obtaining data in a forensically sound manner, addresses difficulties faced in the investigative process, and derives important knowledge from the gathered data.

The outcomes of this research stand to advance I2P forensics, supporting law enforcement agencies, cybersecurity professionals, and forensic practitioners in their quest for pivotal information. The study aids in identifying individuals or entities involved in I2P-related crimes while devising effective investigation and mitigation strategies within the privacy-centric communication networks. By bridging the existing research gap, this study offers a comprehensive analysis of I2P forensics on the Android and macOS platforms, thereby highlighting investigation techniques and artifacts indispensable in addressing challenges stemming from the escalating use of I2P for illicit activities.

The subsequent sections of this research paper are structured as follows: Section 2 highlights related works; Section 3 outlines the methodology employed in this study; Section 4 presents results and discussion; Finally, Section 5 concludes the paper and outlines directions for future research and investigation.

II. Literature Survey

The Numerous authors have investigated into the complexities of the Invisible Internet Project (I2P) network, unearthing critical insights into Windows OS forensics and other anonymity networks. [5] focused their attention on Android and Windows-based devices, conducting forensics by extracting data from ZRAM, RAM, Android Device Bridge logs (ADB), and storage on Android devices. They analyzed artifacts from the registry, RAM, and storage for Windows OS. Their findings showed that Windows 10 OS contained artifacts related to TOR activities, similar to Android 10 OS. However, their study did not extend to TOR forensics for iOS and macOS-based devices. The authors in [6] explored into monitoring the anonymization process of hidden networks within the I2P communication framework. They used data collected from stats.i2p to track communication patterns and recognize limitations tied to behavioral modifications. Their research raised the need for further exploration. Another, research by [7] was conducted to identify and classify TOR traffic across mobile and personal computers. Labeling traffic based on time-related and non-time-related features demonstrated the significance of both variables in identifying and categorizing different traffic levels. [8] explored the interconnection between darknets, focusing on Tor and I2P, using a combined dataset that encompasses both communication networks. The results highlighted the concentration of well-known Tor websites and notable presence of specific domains within the I2P network. [9] conducted a connectivity analysis of the I2P web structure, involving the development of a crawling tool to collect data from eepsites within a controlled virtual environment. This analysis highlighted the decentralized nature of the I2P network.

Another research in [10] focused on the traffic classification of anonymity networks, specifically TOR, I2P, and JonDonym. In their study, the Anon17 public dataset was utilized, and various machine learning classifiers were employed to evaluate the anonymity of these networks and the degree to which their traffic could be identified. In [11], the vulnerabilities and forensic techniques tied to the dark side of I2P were examined. Although the research discussed potential forensic avenues, it did not investigate the actual forensic analysis or data extraction. [12] investigated the impact of shared bandwidth on user anonymity within the I2P network. Their experiments brought to light the potential risks of deanonymization that arise with higher bandwidth allocation. However, their study did not extend to exploring the interplay between the I2P network and TOR, nor did it delve into the specifics of macOS forensics. This leaves a notable gap in understanding the full scope of network interactions and forensic challenges in diverse operating environments. The researchers in [13] concentrated on discovering the impacts of shared bandwidth on I2P network users. Their study utilized three VMs for data collection, aiming to understand the effect of shared bandwidth on user anonymity.

In [14], a forensic assessment was conducted on the unencrypted layer of the TOR network. This involved capturing and analyzing network activity from the TOR Browser within the Tails operating system, demonstrating the possibility of de-anonymizing TOR users by scrutinizing unencrypted headers. In [15], an analysis of ZeroNet sites, a decentralized and censorship-resistant communication network, was carried out using big data techniques. Through experiments, the study showcased the capability to identify dark web categories and specific user behaviors within the ZeroNet environment. [16] proposed an anonymous

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communication system based on Software Defined Architecture, introducing asynchronous communication and file exchange. The system aimed to enhance communication anonymity and minimize correlation between distinct messages. [17] Analyzed various anonymity networks focusing on their varying levels of anonymity, security, and other features. The study primarily involved a comparative analysis of the various networks. In [18], a survey of anonymity services on the dark web, such as Freenet, TOR, and I2P, was conducted. This research focused on understanding the functioning and vulnerabilities of these services. [19] Performed a forensic analysis on Windows 10 browser to trace anonymous attacks on P2P network. The data was collected from the hard disk, registry, and RAM, shedding light on the artifact's suggestive of TOR usage. In [20], the dark net was examined as a platform for criminal activities. Various anonymity services like TOR, I2P, and Freenet were discussed, emphasizing on the need to investigate dark net activities and provide solutions for detection and prevention. The authors in [21] sought to understand the dark web, covering surface, deep, and dark web structures. They conducted a detailed examination of TOR and I2P, including services offered, search engines, and mechanisms for accessing dark web content.

The impact of the dark web on internet governance, focusing primarily on TOR and its anonymity features, was reviewed in [22]. The work emphasized considering illegal activities on these services during investigations. Intelligence mining of illegal underground internet activities, using active and passive node detection methods for data collection, was studied in [23]. The study focused on TOR dark network analysis, gathering personal information-related transactional data. In [24] a book on practical cyber forensics provided insights into collecting, analyzing, and presenting digital forensic findings. This publication proved valuable for understanding macOS forensics. The results uncovered artifacts that could reveal user activities. Real-time network analysis and the TOR forensics help gather artifacts that can provide crucial forensic evidence in crime investigations [26]. The authors in [27] focused on the dark web's notorious involvement in illegal activities like drug trafficking, money laundering, and cybercrime, raising concerns for law enforcement and cybersecurity experts. Their research delved into dark web forensics, involving critical tasks of collecting, analyzing, and preserving digital evidence from this hidden part of the Internet. The authors highlighted challenges tied to anonymity, encryption, and the prevalent use of cryptocurrencies. Additionally, the study extensively explored the current research status in dark web forensics, including methodologies, tools, and persistent challenges faced in this domain.

The literature review reveals a significant gap in I2P forensics, particularly in relation to macOS and Android operating systems. While previous studies have concentrated largely on Windows OS and TOR networks, there's a notable void concerning in-depth forensic analysis on these platforms, which this research aims to address by extracting and examining digital evidence linked to I2P activities on macOS and Android.

III. Methodology

The I2P anonymous network program was installed in Android 11 and Mac OS devices, and different predefined dark web browsing activities were performed on the devices using I2P. As part of data population, many scenarios are created, such as searching for different terms, downloading files, using email services, and visiting e-commerce websites. In order to track data when recovering an artifact, a track of what data is created was kept. Extensive study was done to determine what kind of user traces are left on devices and where they are kept. The RAM dump of the devices was taken and data acquisition was done using forensic tools. The forensic images were examined using Cellebrite UFED, Oxygen forensics, HXD and Magnet Axiom.

TABLE I. Software and Tools used for the Experiment

Software and Hardware	Version
Cellebrite UFED	7.64.0.271
Magnet Axiom	v3.11.0.19007
GuyMager	0.8.13
OSXPmem	3.2
I2P Router software for mac OS	1.9.0
I2P Router software for android	2.2.1
i2P - IRC browser	0.0.1
Mozilla Firefox	113.0.1
mac OS Ventura	13.1
Tecno spark 8T	KG6p

Table I shows the tools that were conFigd to perform the investigation of the I2P anonymous communication. The forensic approach that was employed is described in Fig. 2.

IV. Results And Discussion

The evidence recovered during the analysis of the RAM and hard disk on Mac OS and the file system of Android is presented in the following sections

A. *Android findings*

On Android, the forensic analysis was conducted by analyzing the file system to identify artifacts related to I2P on the device. A keyword search was used to look for the artifacts of interest [30]. Fig. 3 shows the blacklisted hosts in the I2P application on an Android device.

Remnants related to dark web browsing activities on the device were found in several locations. The file system under the folder named "sp (4 files 32 KB)" contains a package for the "net.i2p.android.router.provider.CacheProvider.xml," describing numerous I2P-based activities.

The analysis also identified the websites that were accessed on Android, as shown in Fig. 4. It was observed that minimal traces are left behind on an Android device when the user is using an I2P application.

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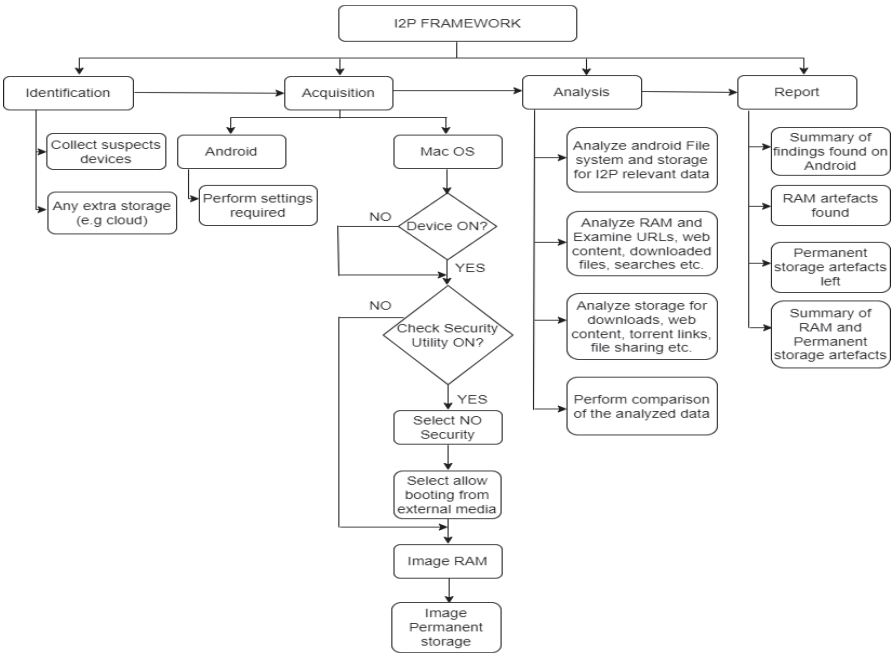

Fig. 2. Framework used for investigation of I2P on Android and mac OS.

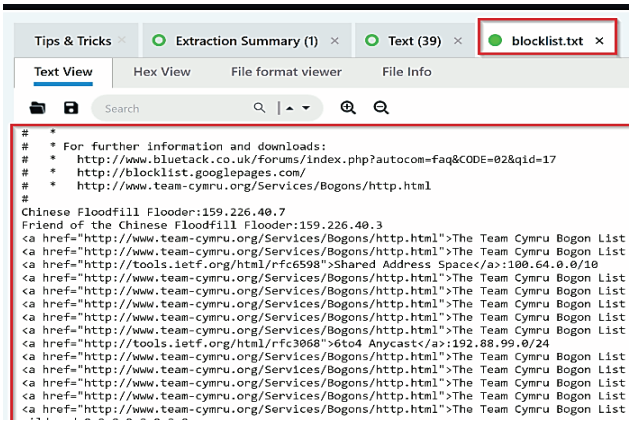

Fig. 3. Blacklisted hosts in I2P application for android.

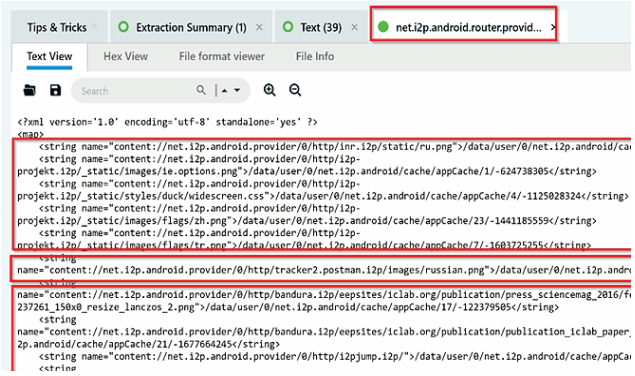

Fig. 4. Eepsites accessed by the user in android.

B. *Mac OS findings*

On Mac OS, forensic analysis was performed on the captured RAM and the hard disk. The following sections address the evidence related to tunnel creation, I2P address book list for eepsites, browsing, bookmarks, mail account registration, downloads, and eepsite web components. RAM analysis identified the tunnel creation related to I2P before communication starts. The tunnels created are shown in Fig. 5.

The analysis of the RAM identified the address book used by the user while using I2P. The address book is the first page accessed by the user when accessing any eepsite. The address book artifacts are shown in Fig. 6

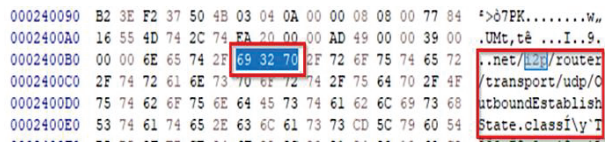

Fig. 5. Tunnel creation process for I2P before communication is established.

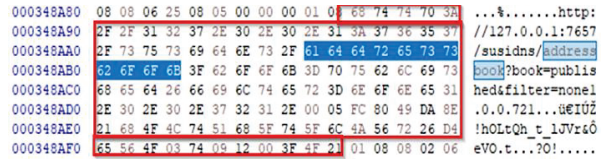

Fig. 6. Address book for the lists of eepsites available by default.

The browsing data for various eepsites bandura.i2p, easygpg2.i2p, inr.i2p, wiki.i2p-projekt.i2p, i2pjump.i2p, identiguy.i2p and others were recovered was recovered, as shown in Fig. 7.

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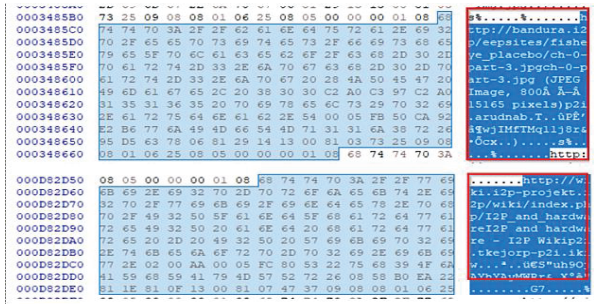

Fig. 7. Browsing related activities.

The analysis identified all the saved bookmarks that were stored under the I2P address book list. Any saved eepsite was saved in the address book, as shown in Fig. 8.

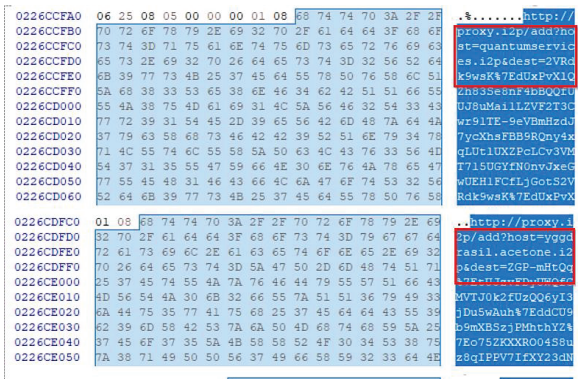

Fig. 8. Bookmarked I2P eepsites.

Artifacts for all the downloads of different files related to I2P were recovered, as illustrated in Fig. 9.

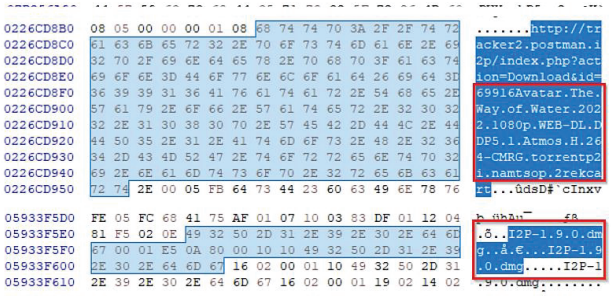

Fig. 9. Downloaded files found.

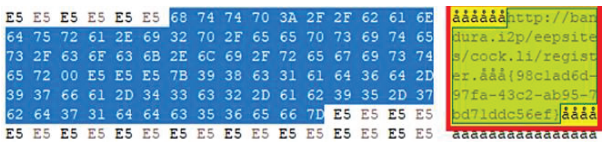

Fig. 10. User account registration artifact

The email registration artifacts were recovered, showing the registration of the email and the eepsite used for creating a new account, as shown in Fig. 10. Further searching did not reveal any other information such as the username details and the password registered for the particular eepsite.

TABLE II. SUMMARY OF THE FINDINGS AND THEIR PRACTICAL IMPLICATIONS

Findings	Practical Implications
Blacklisted hosts in I2P application for Android	Identification of potential security risks and prohibited content within I2P on Android devices
Identification of accessed eepsites on Android	Reconstruction of user activities and connections within the I2P network on Android devices
Remnants of dark web browsing activities	Potential evidence for investigating illicit online activities on Android devices
Tunnel creation process for I2P on Mac OS	Understanding the initial setup and communication processes within the I2P network on Mac OS
Identification of address book artifacts	Reconstruction of eepsite access patterns and user connections within the I2P network on Mac OS
Recovery of browsing related activities	Reconstruction of user browsing history and connections to specific eepsites within the I2P network on Mac OS
Identification of bookmarked eepsites	Insight into user preferences and frequent access patterns within the I2P network on Mac OS
Recovery of downloaded files related to I2P	Potential evidence for understanding user interactions and software usage within the I2P network on Mac OS
Artifacts related to mail account registration	Potential evidence for understanding user communications and account creation within the I2P network on Mac OS

The findings provide valuable insights for forensic investigators, highlighting the potential evidence that can be recovered from both Android and Mac OS environments when conducting forensic analysis related to I2P usage [31]. Table II provides the Forensic Findings for I2P Artifacts on Android and Mac OS and their practical implications.

V. CONCLUSION AND FUTURE WORK

The The forensic analysis of I2P artifacts on Android and Mac OS has revealed valuable insights into the potential evidence that can be recovered from both operating systems. The findings demonstrate the presence of artifacts related to I2P activities, including blacklisted hosts, remnants of dark web browsing activities, accessed eepsites, tunnel creation processes, address book artifacts, browsing history, bookmarks, downloads, and mail account registration. These findings underscore the importance of tailored forensic techniques for different operating systems and the need for ongoing advancements in forensic tools and techniques to effectively investigate I2P-related activities. The research has contributed to advancing forensic analysis techniques in privacy-centric networks and highlights the significance of developing specialized forensic approaches for privacy-centric communication networks such as I2P. By identifying and analyzing these artifacts, forensic practitioners, law enforcement agencies, and cybersecurity professionals can gain valuable insights into user activities within the I2P network, facilitating more efficient investigation of I2P-related crimes [32]. Future work in this area could focus on developing automated tools and methodologies to streamline

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the extraction and analysis of I2P artifacts, particularly in the context of mobile forensics for Android devices. Additionally, further research could explore the integration of machine learning and artificial intelligence techniques to enhance the efficiency and accuracy of I2P forensic analysis.

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